

PROBABILITY

Experiment:

A planned activity or process, whose results yield a set of data, is called an experiment.

Random Experiment:

An experiment which produces different results even though it is repeated a large number of times under similar conditions is called a random experiment i.e. tossing a fair coin, rolling a fair die etc.

Properties of Random Experiment:

A random experiment has the following properties.

1. The experiment can be repeated, practically or theoretically, any number of times.
2. The random experiment always has at least two (two or more than two) possible outcomes.
3. The outcome of each repetition (trial) is unpredictable.

Trial:

A single performance of an experiment is called a trial.

Outcome:

The result obtained from a trial is called an outcome.

Sample Space:

A set consisting of all possible outcomes of a random experiment is called sample space for that experiment and is denoted by 'S'. Each possible outcome of the experiment is a member of sample space and is called **sample point**.

Event:

Any subset of a sample space, describing a particular characteristic, is called an event.

Simple Event:

An event that contains only one sample point is called a simple event. It is also called elementary event.

Compound Event:

An event that contains more than one (at least two) sample point is called compound event.

Impossible Event:

It is an event that contains no sample point. It is denoted by ϕ . It never occurs. Probability of the occurrence of impossible event is zero.

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Sure Event or Certain Event:

It is the event that contains all sample points of the sample space. It is denoted by S . It certainly occurs so probability of the occurrence of sure event is unity or one (1).

Independent Events:

If the occurrence (or non-occurrence) of one event does not affect the occurrence (or non-occurrence) of the other event then these events are called independent events. If A and B are two independent events then

$$P(A \cap B) = P(A) \times P(B) \quad \text{because}$$

$$P(A/B) = P(A) \text{ and } P(B/A) = P(B)$$

$$P\left(\frac{A}{B}\right) =$$

Dependent Events:

If the occurrence (or non-occurrence) of one event affects the occurrence (or non-occurrence) of the other event then the events are called dependent events. If A and B are two dependent events then

$$P(A \cap B) = P(A) \cdot P(B/A) \quad \text{If } A \text{ occurs prior to } B.$$

$$P(A \cap B) = P(B) \cdot P(A/B) \quad \text{If } B \text{ occurs prior to } A.$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Equally Likely Events:

If an event A of a sample space is as likely to occur as event B of the same sample space, then events A and B are said to be equally likely events. This rule can be generalized for any number of events.

Mutually Exclusive Events:

Two events A and B defined on the same sample space are called mutually exclusive events if they cannot occur together. In this case they have no common sample point among them.

OR If the occurrence of an event prevents the occurrence of the other event, then the events are called mutually exclusive events. If ' A ' and ' B ' are mutually exclusive events:

$$\text{then } (A \cap B) = \phi$$

$$\text{and } P(A \cap B) = 0$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Non-Mutually Exclusive Events:

Two events A and B defined on the same sample space are called non-mutually exclusive events if they can occur together or they have at least one sample point in common.

OR If the occurrence of an event does not prevent the occurrence of the other event, then these events are called non-mutually exclusive events. If ' A ' and ' B ' are non-mutually exclusive events:

$$\text{then } (A \cap B) \neq \phi$$

$$\text{and } P(A \cap B) \neq 0$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Exhaustive Events:

If two or more than two mutually exclusive events defined on the same sample space constitute the entire sample space as their union, the events are called exhaustive events.

OR If the union of two or more than two mutually exclusive events defined on the same sample space, generates the whole sample space, the events are called exhaustive events.

DEFINITIONS OF PROBABILITY

1. Mathematical or Classical or Priori Definition of Probability:

If a random experiment can produce 'n' equally likely, mutually exclusive and exhaustive outcomes and out of these outcomes 'm' are considered favourable to the occurrence of a certain event A, then probability of the event A, denoted by $P(A)$, is defined as the ratio $\frac{m}{n}$. Symbolically:

$$P(A) = \frac{\text{Number of favourable outcomes of event A}}{\text{Total number of possible outcomes in sample space}} = \frac{m}{n}$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Limitations: This definition is applicable only when all possible outcomes are equally likely, mutually exclusive and exhaustive.

2. Relative Frequency or Posteriori Definition of Probability:

Keeping in view the limitations of the classical definition, a new approach was introduced to define the probability. "If a random experiment is repeated a large number of times (say n), under the similar conditions and if an event A occurs 'm' times, then the probability of occurrence of event A is defined as the ratio $\frac{m}{n}$ if number of repetitions 'n' becomes very large". Symbolically:

$$P(A) = \lim_{n \rightarrow \infty} \left(\frac{m}{n} \right)$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

Limitation: This definition is applicable only when number of possible outcomes, is infinitely large.

3. The Axiomatic Definition of Probability:

If $A_1, A_2, A_3, \dots, A_k$ are the 'K' events defined on the same sample space 'S' then the real numbers $P(A_1), P(A_2), P(A_3), \dots, P(A_k)$ assigned to the 'K' events respectively are called the probabilities of these events if they satisfy the following axioms:

1. For any event A_i , $P(A_i) \geq 0$, which means that probability of an event cannot be negative.
2. $P(S) = 1$, which means that probability of sure event is unity (one).
3. If A_i and A_j are two mutually exclusive events then:
 $P(A_i \cup A_j) = P(A_i) + P(A_j)$

So from the first two axioms we can say that $0 \leq P(A) \leq 1$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-5327858

General Law of Addition or Addition Law for Non-Mutually Exclusive events:

If 'A' and 'B' are any two events defined on the same sample space S then the probability that at least one of these events occur is given by

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

This law is also known as Additive law of probability or Law of Total Probability.

Law of Addition for Mutually Exclusive Events:

If A and B are two mutually exclusive events defined on the same sample space, then
Then $(A \cap B) = \phi$
and $P(A \cap B) = 0$
Hence the probability that A or B occurs is given by:
 $P(A \cup B) = P(A) + P(B)$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-3327836

Conditional Probability:

In practical life, so many situations are there when some additional information is given about the outcomes of the experiments. In these cases we consider only those outcomes which are favoring a certain event and excluding some outcomes as being impossible, which were believed to be included in the sample space before receiving that additional information. It means that the actual sample space is reduced due to the additional information about the outcomes of the experiment. This kind of probability which is associated with the reduced sample space is called **Conditional Probability**.

OR If A and B are two dependent events defined on the same sample space, then probability of the occurrence of the event A when it is known that event B has already occurred is called conditional probability of event A when event B has already occurred and is denoted by $P(A/B)$. It is also read as probability of 'A given B'.

Conditional probability of an event is given by:

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

If event 'B' occurs prior to event 'A'.

$$P(B/A) = \frac{P(A \cap B)}{P(A)}$$

If event 'A' occurs prior to event 'B'.

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-3327836

Multiplication Law for Dependent Events:

If 'A' and 'B' are two dependent events defined on the same sample space, then probability that both of them occur is the product of the probability of event A and conditional probability of event B given A; if A occurs prior to event B. Symbolically:

$$P(A \cap B) = P(A) \cdot P(B/A) \quad \text{if A occurs prior to B.}$$

$$P(A \cap B) = P(B) \cdot P(A/B) \quad \text{if B occurs prior to A.}$$

Multiplication Law for Independent Events:

If A and B are two independent events then occurrence (or non-occurrence) of A does not affect the occurrence (or non-occurrence) of B and vice versa. Thus if A and B are independent events then:

$$P(A/B) = P(A)$$

$$P(B/A) = P(B)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

OR If A and B are two independent events then the probability that both of them occur is equal to the product of their individual probabilities. Symbolically:

$$P(A \cap B) = P(A) \cdot P(B)$$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt.
Cell: 0333-3327836

RANDOM VARIABLE AND PROBABILITY DISTRIBUTION

Random Variable:

A variable whose values are determined by the outcomes of a random experiment is called a random variable. A random variable is also called chance variable or stochastic variable.

Discrete Random Variable:

A random variable which takes on only a finite number of values or a sequence of whole numbers is called a discrete random variable. A discrete random variable is usually characterized by counting process. Number of students in different class rooms of a college, Number of accidents on motorway per day and Number of children per family are different examples of discrete random variable.

Continuous Random Variable:

A random variable which can assume an infinite number of values in a given interval is called continuous random variable. It is to be noted that continuous random variable is characterized by measurement process i.e. heights, weights, time, temperature, distances and lifetimes etc.

Discrete Probability Distribution:

The tabular arrangement of all possible values of a discrete random variable 'X' i.e. $x_1, x_2, \dots, x_n$ along with their respective probabilities $f(x_1), f(x_2), \dots, f(x_n)$ is called probability distribution. It has following properties:

- $0 \leq f(x_i) \leq 1$ i.e. Probability lies between 0 and 1, both inclusive.
- $\sum_{i=1}^n f(x_i) = 1$ i.e. The sum of probabilities is unity (1).

Probability Function:

If the function or formula $f(x)$ generates the probabilities of a discrete random variable 'X', it is called probability function. It possesses the following two properties:

- $0 \leq f(x_i) \leq 1$ i.e. Function is non-negative for the given value of 'x'.
Probability lies between 0 and 1, both inclusive.
- $\sum_{i=1}^n f(x_i) = 1$ i.e. The sum of probabilities is unity (1).

Probability Density Function:

The probability distribution of a continuous random variable cannot be presented in tabular form because it can assume infinite number of values. It can be represented by means of a formula or a graph. A function $f(x)$ is called a probability density function (p.d.f.) for the continuous variable 'X' if it is non negative and the total area under its curve and above the x-axis is equal to one and the area under the curve between two values $x = a$ and $x = b$ gives the probability that 'X' lies between 'a' and 'b'. Thus p.d.f. has the following properties:

- $f(x) \geq 0$ i.e. function is non-negative.
- $\int_{-\infty}^{\infty} f(x) dx = 1$ i.e. total area under the curve and above x-axis is unity (1).

$$P(a \leq x \leq b) = \int_a^b f(x) dx = \text{same}$$

$$P(a < x < b) = P(a \leq x \leq b) = P(a < x < b)$$

No. of accidents on M2 (motorway) in a month

Compiled by: Saif-ur-Rehman, Assistant Professor, Department of Statistics, F. G. Sir Syed College, The Mall, Rawalpindi. Mobile: 0333-5327858

9 of 43

Quantitative variable / Attribute

- marks in SSC

- marks in Maths

- No. of passing subjects
- Family size

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt
Cell: 0333-5327858

Waist size is discrete variable but it is measurement

X	f(x)
x_1	$f(x_1)$
x_2	$f(x_2)$

SAIF-UR-REHMAN
Assistant Professor
F.G. Sir Syed College
Rawalpindi Cantt
Cell: 0333-5327858

function = Tossing 4 coins Prob of head =
i.e. The sum of probabilities is unity (1).
 $P(X=x) = {}^4C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{4-x}$ $\therefore x = 0, 1, 2, 3, 4$

Distribution Function: It is basically cumulative frequency of probability

A function showing probabilities that a random variable 'X' has a value less than or equal to 'x' is called cumulative distribution function of 'X' or Distribution function of 'X', and denoted by $F(x)$. Mathematically: $F(x) = P(X \leq x)$

The distribution function has the following properties.

1. $F(-\infty) = 0$ and $F(\infty) = 1$

2. If $a < b$ then $F(a) < F(b)$, where 'a' and 'b' are real numbers.

The distribution function of a continuous random variable 'X' is defined as:

$$F(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx = P(-\infty < X \leq x)$$

Mathematical Expectation or Expected Value of a Random Variable:

Let a discrete random variable 'X' has the values $x_1, x_2, \dots, x_n$ with corresponding probabilities $f(x_1), f(x_2), \dots, f(x_n)$ such that $\sum_{i=1}^n f(x_i) = 1$. The mathematical expectation of 'X' or Expected value of 'X' is the arithmetic mean of the discrete random variable 'X' is denoted by $E(X)$ and defined as:

$$E(X) = x_1 f(x_1) + x_2 f(x_2) + \dots + x_n f(x_n)$$

$$E(X) = \sum_{i=1}^n (x_i f(x_i))$$

The expectation of a continuous random variable is given by:

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx \quad E(C) = C$$

Laws of Expectation or Properties of Expectation:

- $E(C) = C$ Expectation of a constant is the constant itself.
- $E(aX + b) = aE(X) + b$
 $E(aX) = aE(X)$
 $E(X + b) = E(X) + b$
- $E(X + Y) = E(X) + E(Y)$
 $E(X - Y) = E(X) - E(Y)$
- $E(XY) = E(X) \cdot E(Y)$ if X and Y are independent to each other.
- $E[X - E(X)] = 0$ Sum of deviations of expected value is zero
- $E[X - E(X)]^2 = \text{Var}(X)$

Variance and Standard Deviation:

The variance of a random variable 'X', denoted by $\text{Var}(X)$ or σ^2 , is defined as:

$$\text{Var}(X) = \sigma^2 = E[X - E(X)]^2 = [E(X^2) - (E(X))^2]$$

$$S.D(X) = \sigma = \sqrt{E[X - E(X)]^2} = \sqrt{E(X^2) - (E(X))^2}$$

Properties of Variance

- $\text{Var}(K) = 0$
- $\text{Var}(X \pm b) = \text{Var}(X)$
- $\text{Var}(aX) = a^2 \text{Var}(X)$
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$
- $\text{Var}(X \pm Y) = [\text{Var}(X) + \text{Var}(Y)]$
 If X and Y are independent to each other.
- $E[X - E(X)]^2 = \text{Var}(X)$

Properties of Standard Deviation

- $SD(K) = 0$
- $SD(X \pm b) = SD(X)$
- $SD(aX) = |a| SD(X)$
- $SD(aX + b) = |a| SD(X)$
- $SD(X \pm Y) = \sqrt{\text{Var}(X) + \text{Var}(Y)}$
 If X and Y are independent to each other.
- $\sqrt{E[X - E(X)]^2} = SD(X)$